

Experiment - 10

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Semester: 5	Date of Performance: 14/9/26
Subject Name: ADBMS	

(A)

Design a robust PostgreSQL transaction system for the students table where multiple student records are inserted in a single transaction. If any insert fails due to invalid data, only that insert should be rolled back while preserving the previous successful inserts using savepoints. The system should provide clear messages for both successful and failed insertions, ensuring data integrity and controlled error handling.

Solution:

```
CREATE OR REPLACE PROCEDURE insert_students()
LANGUAGE PLPGSQL
AS $$
BEGIN
    BEGIN
        INSERT INTO students(name, age, class)
        VALUES ('Anisha', 16, 8);

        RAISE NOTICE 'Anisha inserted successfully';

    EXCEPTION WHEN OTHERS THEN
        RAISE NOTICE 'Anisha insertion failed. Rolled back.';
    END;
END;
```

```
INSERT INTO students(name, age, class)
VALUES ('Neha', 17, 8);
```

```
RAISE NOTICE 'Neha inserted successfully'; EXCEPTION
```

```
WHEN OTHERS THEN
```

```
RAISE NOTICE 'Neha insertion failed. Rolled back.';
END;
```

```
INVALID BEGIN
```

```
INSERT INTO students(name, age, class)
VALUES ('Anisha', 19, 9);
```

```
RAISE NOTICE 'Anisha inserted successfully';
```

```
EXCEPTION WHEN OTHERS THEN
```

```
RAISE NOTICE 'Anisha insertion failed. Only this insertion was rolled
back.'; END;
```

```
BEGIN
```

```
INSERT INTO students(name, age, class)
VALUES ('Mayank', 19, 9);
```

```
RAISE NOTICE 'Mayank inserted successfully';
```

```
EXCEPTION WHEN OTHERS THEN
```

```
RAISE NOTICE 'Mayank insertion failed. Rolled back.';
END;
```

```
RAISE NOTICE 'Transaction Successfully Done'; END;
$$;
```

```
BEGIN;
```

```
CALL insert_students();
```

```
COMMIT;
```

```
SELECT * FROM students;
```